Code No. 8410

OSMANIA UNIVERSITY
FACULTY OF ENGINEERING
UNIVERSITY COLLEGE OF ENGINEERING (AUTONOMOUS)
B.E. (AI&ML) IV-Semester (MAIN) Examinations July-2025

100523729008

SOFTWARE ENGINEERING

Time : 3 hours **Max. Marks : 60**

Note : *i) Answer **Question No. 1** (Compulsory) and answer any **four questions** from the remaining questions (2- 7). Each question caries 12 marks.*
ii) Answers must be written in same order as they occur in the Question Paper.
*iii) **Missing data, if any may suitably be assumed.***

		Marks	BT	CO
1. a)	Mention the benefits of UML diagrams in the process of software product development.	2	1	1
b)	List disadvantages of the Incremental process model.	2	2	2
c)	What is the Key difference between UML Diagram and Data Flow Diagram?	2	2	3
d)	What are the Golden rules applicable to the design of User Interface?	2	2	4
e)	Mention the benefits of Regression Testing.	2	1	5
f)	How is the Black-box Testing different than White-box Testing?	2	2	5
2. a)	A Rectangle class and a Circle class implement the Shape interface, which declares a getArea() method. The Rectangle is defined with properties, length and width while Circle is with radius. Draw the class diagrams using the modelling language, UML.	6	3	1
b)	Summarize the advantages and disadvantages of Agile methodology.	6	1	1
3. a)	Describe precisely about the Construction Principles applicable to the s/w product development.	6	3	2
b)	Discuss the ways how to address the conflicts raised in approving the Requirements document of a software product.	6	4	2
4. a)	The food ordering system has four entities; Customer, Supplier, Kitchen and Manager and these interact with the system. Build the data flow diagram, **Level 0**, representing the entities how data flows.	6	4	3

b)	What is Architecture Trade off Analysis Method (ATAM) used for? Describe the steps that follow in sequence that follows the design analyses using ATAM.	6	3	3
5. a)	Draw the UML diagram and then using the Object Constraint Language, represent the condition of "A vehicle owner must be at least 18 years old" from the given oclType s, Person and Vehicle; Car and MotorCycle are the sub types of Vehicle. A Person has attributes, name and age. A Vehicle has attribute, Color.	6	5	4
b)	What is Cohesion and Coupling? Why does the high Cohesion and low Coupling are highly desirable in the software design and development?	6	4	4
6. a)	Explain what is debugging process as identified one of the processes in the Software Development. How does the process is engaged in testing the product?	6	3	5
b)	What is Unit testing? Describe the testing frameworks (tools) for testing code quality applicable to any four of the coding languages.	6	3	5
7. a)	*Illustrate* with examples of good characteristics of User Interface design.	6	1	4
b)	Summarize the Process and Product metrics for assessing the quantitative measurements of Testing the software product.	6	3	5

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